Signal2Stress

From News to Portfolio Risk Intelligence

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S&P Global x Crisil Hackathon - Phase 3



The Risk Intelligence Gap

“The gap delays confident, portfolio-specific risk decisions.”

Manual Bottleneck

Portfolio managers spend 60% of their day filtering news noise. Manual review is too slow for breaking events, leading to delayed responses in high-volatility environments.

Hallucination Risk

Standard AI models often misinterpret financial syntax or hallucinate metrics. In risk management, a 1% error in sentiment extraction can lead to catastrophic mispricing.

From News to Stress

End-to-end pipeline that turns unstructured news into portfolio stress intelligence

01
News intake

Continuous ingest from global news and filings

02
Signal extraction

LLMs identify events, entities and sentiment

03
Exposure mapping

Map signals to sectors, assets and portfolios

04
Asset-specific stress

Compute impact across scenarios and horizons

05
Dashboard + API ledger

Actionable views, APIs and an audit trail for every step







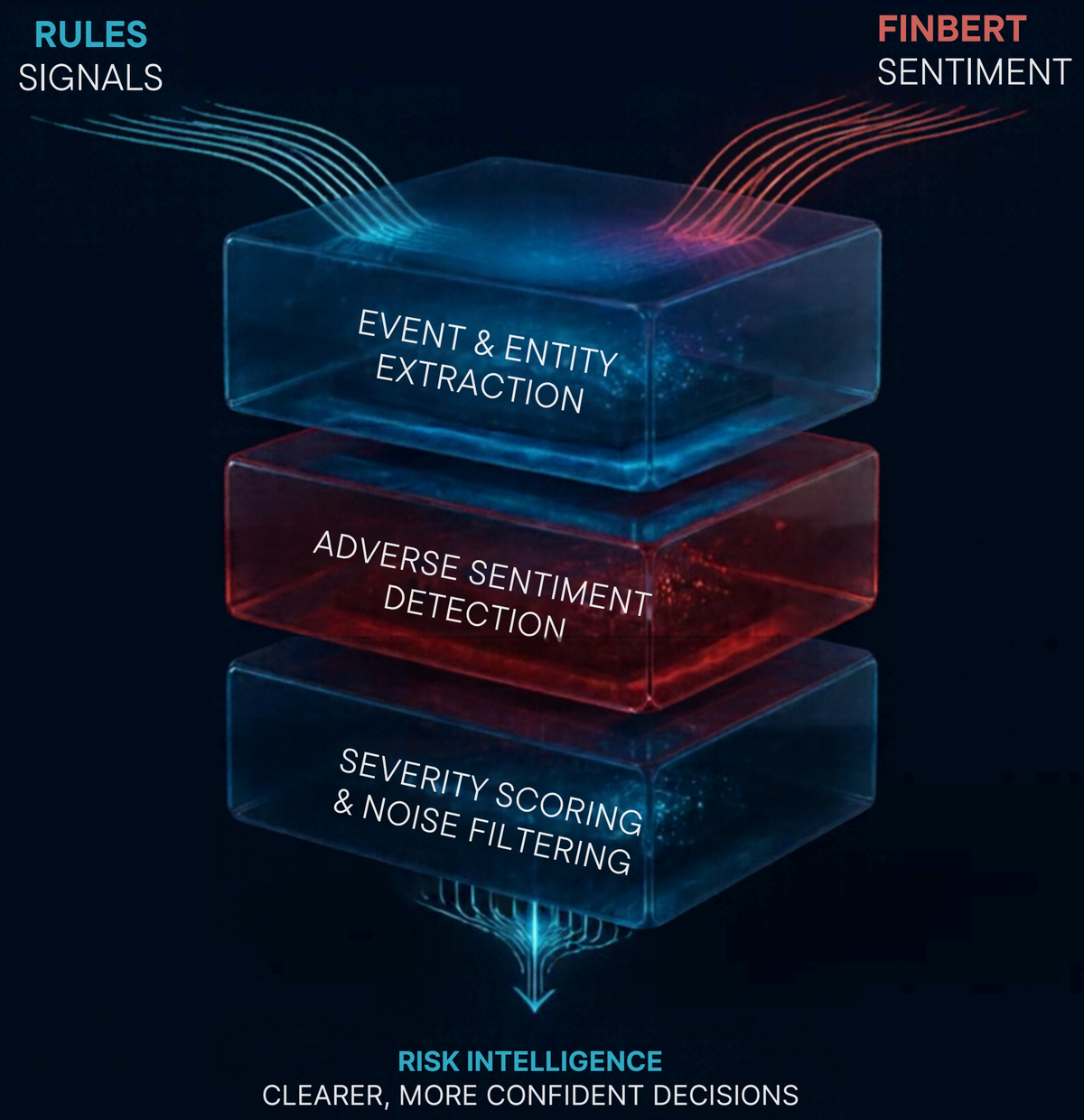
AI/NLP Risk Engine

Regex for explicit
events and entities

FinBERT
adverse sentiment

Severity scoring
and noise filtering

Complementary
signals, stronger decisions



Mapping & Stress Engines

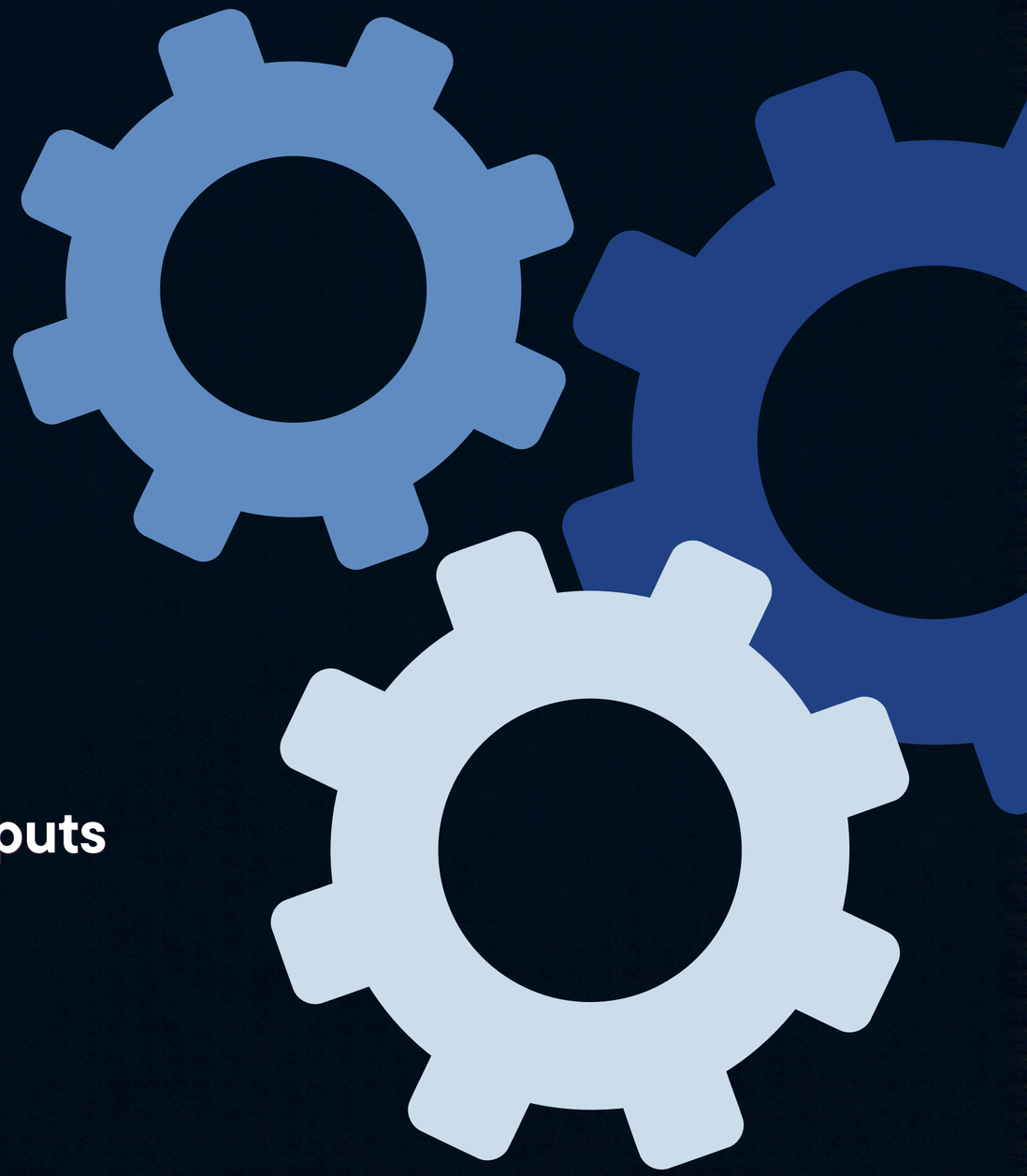
DIRECT
firm-level exposures

→

SECTOR
industry & peer drivers

→

MACRO
rates, credit & market shocks



- **FIRST-ORDER ASSET-SPECIFIC STRESS**
- **Pydantic contracts power clean, validated inputs**
- **Loans stress flows via ECL impact**
- **Bonds stress prices through duration shock**
- **Swaps stress P&L via DV01 shift**

Demonstration

Streamlit app.py

live dashboard experience

Interactive news, GDELT, CSV

explore, filter, and analyze

FastAPI api.py

documented endpoints

SQLite audit ledger

immutable risk decision trail



Results & Next Steps

“Milliseconds to a decision, with a clear path to production.”

Result: reproducible risk attribution from unstructured news without depending on expensive LLM calls.

Impact: faster, more explainable decisions during systemic stress events.

Next: real portfolio integration, advanced propagation models, and live Bloomberg Terminal connectivity.